A Battery State of Health Prediction Framework Considering User Behavior for On-road Electric Vehicles

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***Abstract*—This paper proposes a battery state of health (SOH) prediction framework based on a field dataset of 200 on-road electric vehicles collected during June 2022 to June 2023. 21 user charging and driving behavior features are extracted from the field data. A long short term memory (LSTM) neural network is employed to predict the battery SOH. The impacts of the 21 features on the battery SOH degradation are ranked.**

***Index Terms*—On-road electric vehicles, battery state of health, user behavior, field data, long short term memory.**

I. Introduction

To enhance the system safety and economy, the estimation of the state of health (SOH) of lithium-ion batteries used in electrified transportation systems and renewable energy systems is of great interest. Various methods have been proposed to characterize and estimate the battery SOH, which are mainly based on laboratory data collected from specifically designed experiments [1]–[3]. While such methods are effective in estimating the battery SOH in certain application scenarios, in general they cannot be readily applied to real-world systems as the battery operating conditions may vary significantly from application to application.

For electric vehicle (EV) applications, the battery usage patterns vary from vehicle to vehicle, which may lead to different battery degradation patterns. In fact, the EV user behavior has been investigated. The user driving style can be characterized by specific driving actions such as speed and acceleration [4]–[6]. The effects of the EV user behavior on the battery degradation process have also been explored [7]–[9]. For instance, the effects of the resting behavior (e.g., proportion of the resting time with respect to the usage time) and the charging behavior of the user on the battery capacity degradation are analyzed [7]. As another example, the fast charging behavior of the user is considered to estimate the battery capacity [8].

To further investigate the impacts of the EV user behavior on the battery SOH degradation, this paper proposes a battery SOH prediction framework based on a field dataset collected by the Shanghai Electric Vehicle Public Data Collecting, Monitoring, and Research Center (EVData) located in Shanghai, China [10]. 21 user charging and driving behavior features are extracted from the field data. A long short term memory (LSTM) neural network is utilized to predict the SOH on a monthly basis. The impacts of the 21 features on the SOH degradation are ranked.

TABLE I
EV Models and Battery Characteristics

EV model	Battery chemistry	Battery energy (kWh)	Battery voltage (V)	Battery capacity (Ah)
123114716	NCM	50.8	350	145
123114912	NCM	50.8	350	145
123114718	NCM	52.5	350	150
123115361	NCM	61.1	351	174
123114323	NCM	35.0	350	100
123115359	LFP	50.3	384	131

The remainder of this paper is organized as follows. Section II illustrates the battery field data. Section III presents the battery SOH prediction framework. Section IV evaluates the proposed framework. Section V concludes this paper.

II. Battery Field Data

The battery field data of 200 passenger EVs of six sedan models is utilized in this paper. Table I lists the key characteristics of the lithium-ion batteries used in these EVs. Note that the EV models are denoted by the sequence numbers as the actual manufacturers and model names are confidential. The field data was collected during the period of June 2022 to June 2023. Among the six sedan models, five models adopt the NCM chemistry and the remaining one is based on the LFP chemistry. The battery capacity is between 100 and 174 Ah. The charging and driving data associated with the 200 EVs is collected, which is used to specify the charging and driving behaviors of the EV users. The key items of the dataset are listed in Table II.

III. Battery SOH Prediction Framework

A. Overview

The proposed battery SOH prediction framework is illustrated in Fig. 1. Based on the EV charging data (e.g., charging current and SOC), the battery SOH is determined and labeled. Meanwhile, the charging behavior features such as the number of charging segments per month are extracted. On the other hand, based on the driving data (e.g., speed and mileage), the

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TABLE II
KEY ITEMS OF BATTERY FIELD DATA

Items	Remarks
collectiontime	Data collection time
vehicledata_chargestatus	Vehicle charging status
vehicledata_speed	Vehicle speed in km/h
vehicledata_summileage	Vehicle total mileage in km
vehicledata_sumcurrent	Battery total current in A
vehicledata_soc	Battery SOC from 0 to 100
vehicledata_insulationresistance	Insulation resistance in kΩ
vehicledata_acceleratorpedal	Accelerator pedal from 0 to 100
alarmdata_comalarmflag	Alarm flag

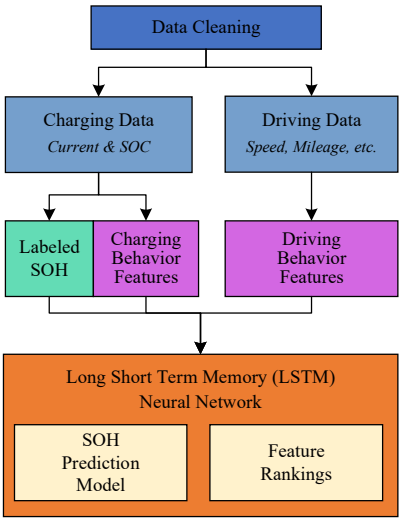


Fig. 1. Battery SOH prediction framework.

driving behavior features such as the average speed per month are extracted. The SOH labels together with the extracted features are fed into a LSTM neural network that generates a SOH prediction model. Moreover, the impacts of the features on the SOH degradation are analyzed and ranked.

B. Battery SOH Labeling Process

To label the battery SOH, the battery capacity needs to be determined first. A variant of the Coulomb counting method originally developed for battery SOC estimation is utilized to calculate the battery capacity Q:

$$Q = \frac{\Sigma I(t) \cdot \Delta t}{\Delta SOC} \tag{1}$$

where Δt is the sampling interval, $I(t)$ is the charging current, and ΔSOC is the change in the SOC after the charging process. One of the issues with this method is that to obtain an accurate estimate of the capacity, an accurate estimate of the SOC and a relatively large SOC change are required, which are not always available, especially in field datasets [11].

To solve this problem, the battery capacity is carefully estimated and the battery SOH labeling process is fine tuned, as illustrated in Fig. 2. First, smooth all the charging segments by

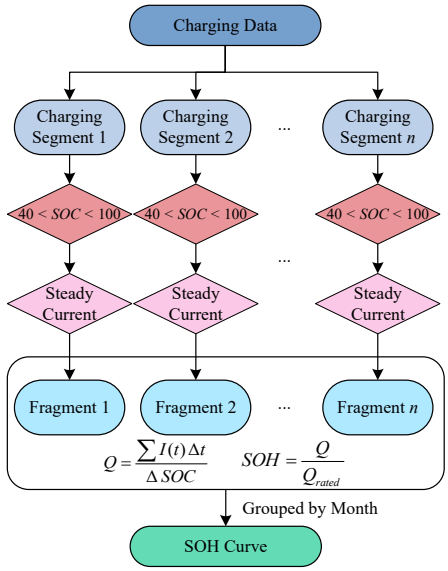


Fig. 2. Battery SOH labeling process.

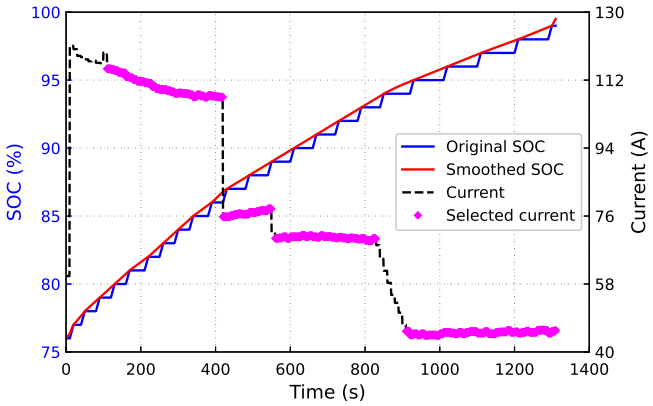


Fig. 3. Charging process screening.

changing the sampling interval from 10 to 1 s by interpolating. As illustrated in Fig. 3, the original staircase-shaped SOC curve is smoothed and the resolution of the smoothed SOC curve is significantly improved compared to that of the original SOC curve.

Second, select the charging segments with SOCs between 40% and 100%. The charging segments with SOCs lower than 40% are not used to determine the battery capacity because the estimation errors of lower SOCs are usually larger.

Third, within the selected charging segments, the fragments with steady currents (i.e., currents that do not change abruptly) are picked. More specifically, if the difference between the magnitudes of two adjacent samples of the current is no more than 1 A, the current is considered steady and the corresponding fragment can be used to estimate the battery capacity. This screening process removes the fragments with abruptly changing currents that may lead to significant errors in estimating the battery capacity as the current is part of

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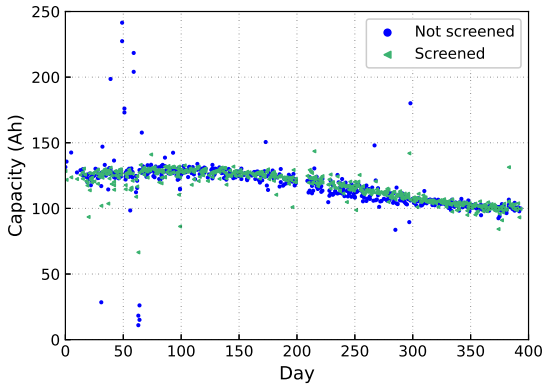


Fig. 4. Estimated capacities before and after charging process screening.

the numerator of (1). As illustrated in Fig. 3, around 420 s, the current jumps from approximately 94 to 85 A. If a constant current is used in (1) to estimate the battery capacity, a significant error will be introduced. Therefore, the fragment closely around 420 s is excluded.

To illustrate the effects of charging process screening, Fig. 4 shows the estimated capacities for a particular battery before and after screening the charging process using the procedure above. Clearly, the dispersion of the capacities estimated using the selected fragments is much lower.

Finally, the battery SOH is calculated with respect to the rated capacity:

$$SOH = \frac{Q}{Q_{rated}} \tag{2}$$

For each month, the SOHs determined using all the selected fragments are averaged and aggregated as a single label.

C. User Behavior Feature Extraction

To investigate the impacts of the EV user behavior on the battery SOH degradation, 21 features characterizing the user charging and driving behavior are extracted from the charging and driving data, as listed in Table III. Note that these features are extracted on a monthly basis.

To characterize the user charging behavior, eight features are extracted from the charging data. While most features are self-explanatory, two of them are elaborated as they are related to the user fast-charging behavior. First, the "Ave_max_charge_current" feature averages the maximum charging current in each charging segment, which is a measure of the charging rate. Second, the "Fast_charge_rate" feature explicitly measures the user fast-charging behavior, which is determined as

$$FC = \frac{N_{FC}}{N_{charge}} \tag{3}$$

where N_{FC} is the number of the charging segments whose maximum charging rates are over 1 C and N_{charge} is the number of all the charging segments.

For the user driving behavior, 13 features are extracted and some of them are elaborated as follows. First, three speed features are extracted and one of them is used to

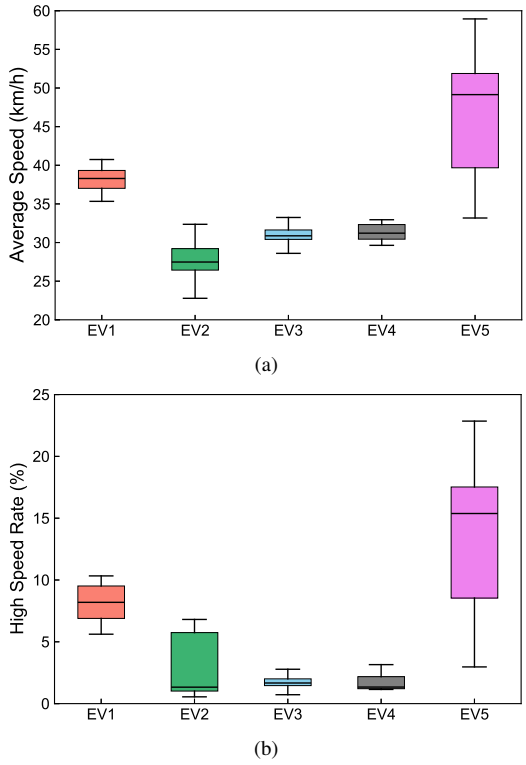


Fig. 5. Speed features of five EVs of model 123114912. (a) Speed_ave. (b) High_speed_rate.

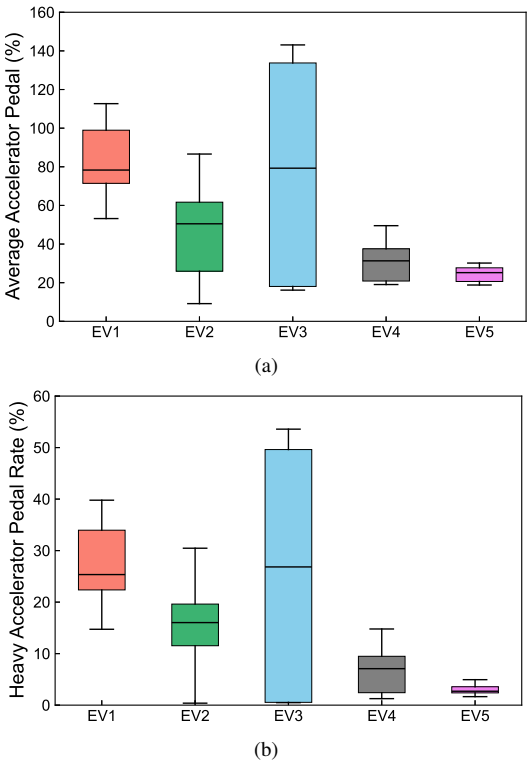


Fig. 6. Accelerator pedal features of five EVs of model 123114912. (a) Acc_ped_ave. (b) Heavy_acc_ped_rate.

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TABLE III
USER BEHAVIOR FEATURES

	Features	Remarks (Features are extracted on a monthly basis)
Charging behavior features	Charge_segment_num	Number of charging segments
	Ave_charge_current	Average of charging currents in charging segments
	Ave_max_charge_current	Average of maximum charging currents in charging segments
	Ave_charge_time	Average of charging time (in seconds) in charging segments
	Ave_charge_start_soc	Average of initial SOCs when charging segments start
	Ave_charge_end_soc	Average of final SOCs when charging segments end
	Ave_charge_soc_change	Average of SOC changes during charging segments
	Fast_charge_rate	Rate of fast-charging segments
Driving behavior features	Speed_ave	Average speed
	Speed_std	Standard deviation of speed
	High_speed_rate	Rate of high-speed driving
	Insulation_res_ave	Average insulation resistance
	Insulation_res_std	Standard deviation of insulation resistance
	Insulation_res_abnor_rate	Rate of abnormal insulation resistance
	Acc_ped_ave	Average accelerator pedal
	Heavy_acc_ped_rate	Rate of heavy accelerator pedal
	Alarm_times	Number of alarms
	Delta_mile	Monthly mileage
	Sum_mile	Total mileage
	Capacity	Battery capacity
	Month	Month

measure the rate of high-speed driving, which is defined as the "High_speed_rate" feature and calculated as

$$HS = \frac{N_{HS}}{N_{drive}} \tag{4}$$

where N_{drive} is the number of all the sampling points in the driving data and N_{HS} is the number of the sampling points with speeds higher than 80 km/h in the driving data. This feature is extracted to complement another two features related to the EV speed: "Speed_ave" for the average speed and "Speed_std" for the standard deviation of the speed. Fig. 5 shows the speed features of five EVs of model 123114912 during the period of June 2022 to June 2023: Fig. 5(a) for the average speed and Fig. 5(b) for the high-speed rate. Clearly, the user behaviors vary: EV5 and EV2 with the highest and the lowest "Speed_ave"/"High_speed_rate", respectively.

Second, two features are extracted to characterize the accelerator pedal behavior: "Acc_ped_ave" for the average accelerator pedal and "Heavy_acc_ped_rate" for the rate of the heavy accelerator pedal. Note that the accelerator pedal is denoted by a number between 0 and 100, as listed in Table II. The pedal is considered "heavy" if its value is greater than or equal to 60. Therefore,

$$HP = \frac{N_{HP}}{N_{drive}} \tag{5}$$

where N_{HP} is the number of the sampling points with heavy accelerator pedals and N_{drive} is the same as in (4). For instance, Fig. 6 shows the two pedal features of the five EVs examined in Fig. 5. EV3 is the highest in terms of both features and EV5 is the lowest. Combined with Fig. 5, it can be assumed that while the EV5 driver is relatively aggressive in terms of the three speed features, this driver is rather conservative in terms of the two pedal features. Therefore, if jointly used, these features help characterize the EV user behavior in a more comprehensive manner.

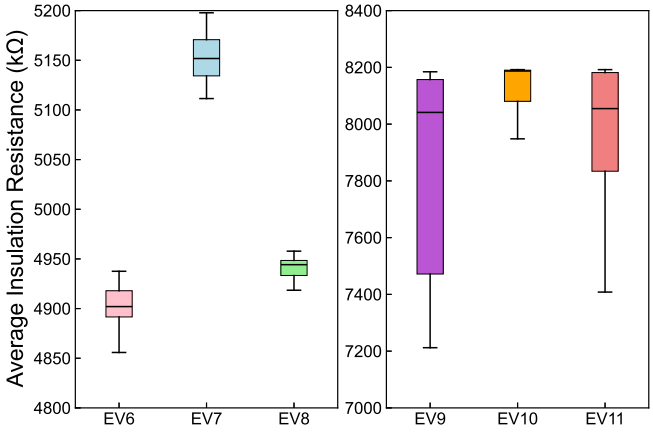


Fig. 7. Average insulation resistances of six EVs: EV6-EV8 of model 123114716 and EV9-EV11 of model 123114912.

Finally, three features are extracted to characterize the EV insulation resistance: "Insulation_res_ave" for the average insulation resistance, "Insulation_res_std" for the standard deviation of the insulation resistance, and "Insulation_res_abnor_rate" for the rate of the "abnormal" insulation resistance. Fig. 7 shows the average insulation resistances of six EVs: EV6, EV7, and EV8 are of model 123114716 while EV9, EV10, and EV11 are of model 123114912. This feature

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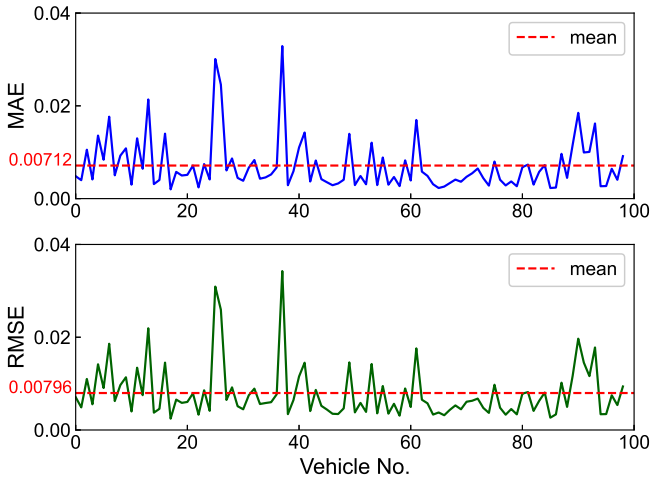


Fig. 8. Errors of EVs in training set in best trial.

varies with EV and model. For EV6-EV8 of the same model, their average insulation resistances range between 4800 and 5200 kΩ. For EV9-EV11 of another model, their resistances are higher: 7200-8200 kΩ. For all the six models, if the average insulation resistance of a particular EV is less than 4000 kΩ or greater than 7500 kΩ, it is considered "abnormal" and taken into account in the "Insulation_res_abnor_rate" feature.

IV. Evaluation of Proposed Framework

A. Evaluation Setup

As illustrated in Fig. 1, a LSTM neural network is employed to predict the battery SOH. Specifically, based on the 21 features and the SOH for the present month, the LSTM model predicts the SOH for the next month. The LSTM model is composed of two layers. The first layer has 256 nodes and the second layer has 512 nodes. The "Adam" optimizer is used in the training process. The learning rate is set as 1e-5 and the dropout rate is set as 0.2 to avoid over-fitting.

Among the 200 EVs, 99 of them with sufficiently long records of data (i.e., at least 10 months) are used to evaluate the proposed framework. Based on the field data of these EVs, a six-fold cross validation is performed. Specifically, the 99 EVs are divided into six mutually exclusive folds and each trial utilizes 93 EVs for training and six EVs (approximately 6% of the 99 EVs) for testing. Note that the six EVs in the testing set vary from trial to trial.

B. Evaluation Results: SOH Prediction Errors

The accuracy of the SOH prediction is evaluated using two metrics: the mean absolute error (MAE) and the root mean square error (RMSE), which are determined as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{6}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{7}$$

TABLE IV
SOH Prediction Errors of Six EVs in Testing Set

	EVT1	EVT2	EVT3	EVT4	EVT5	EVT6
MAE	0.0145	0.0054	0.0113	0.0072	0.0082	0.0078
RMSE	0.0179	0.0078	0.0138	0.0084	0.0101	0.0095

where y_i is the true SOH, $\hat{y}_i$ is the predicted SOH, and n is the number of samples.

For each trial, the error of each of the 93 EVs in the training set is recorded. The average MAE and RMSE of all the 93 EVs are calculated. For all the six trials, the maximum average MAE and RMSE are 0.00877 and 0.00906, respectively. The trial with the minimum average MAE and RMSE is designated as the "best" trial and the corresponding errors are shown in Fig. 8. The maximum MAE and RMSE of all the 93 EVs are both below 0.04. The average MAE and RMSE are 0.00712 and 0.00796, respectively.

The SOH prediction results for the testing set in the best trial are extracted. Fig. 9 shows the true and predicted SOH patterns of the six EVs in the testing set, which are of three different models. The vertical red dashed lines indicate the months from which the SOHs are predicted. In general, the predicted SOH patterns follow the true SOH patterns although the deviations are relatively large for certain months. For instance, in Fig. 9(a), the true and predicted SOHs for November 2022 are approximately 0.885 and 0.855, respectively, which leads to an error of about 0.03. For all the six EVs in the testing set, their errors are listed in Table IV. The maximum error is observed for EVT1: 0.0145 for MAE and 0.0179 for RMSE. The average error of all the six EVs in the testing set is 0.0091 for MAE and 0.0112 for RMSE, respectively. Therefore, the proposed framework can be regarded effective in predicting the battery SOH.

C. Evaluation Results: Feature Impacts

The impacts of the 21 features on the SOH degradation are examined by calculating the perturbation loss [12]. The rankings of the feature impacts are listed in Table V. The three highest ranked features are the average accelerator pedal, the average insulation resistance, and the standard deviation of the insulation resistance.

V. Conclusion

This paper proposes a battery SOH prediction framework based on a field dataset. 21 user charging and driving behavior features are extracted from the field data. A LSTM neural network takes the battery SOH of the present month and the 21 features as the input and predicts the SOH of the next month. When the prediction framework is properly tuned (i.e., trained in the best trial), the maximum average MAE and RMSE for an EV in the testing set are less than 1.8% (i.e., RMSE of 0.0179 for EVT1 in Table IV) during the period of June 2022 to June 2023. The impacts of the 21 features on the battery SOH degradation are ranked and the average accelerator pedal turns out to be the highest ranked feature.

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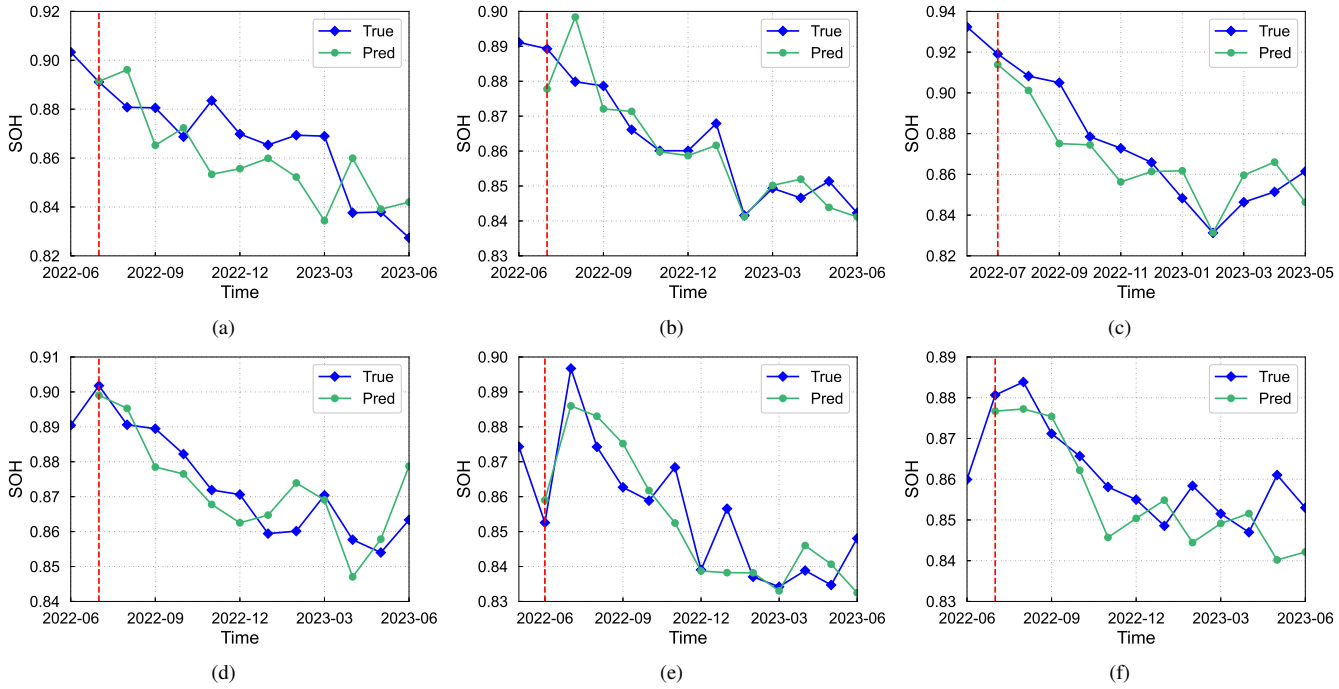


Fig. 9. SOH prediction results of six EVs in testing set. (a) One EV of model 123114323. (b) and (c) Two EVs of model 123114716. (d), (e), and (f) Three EVs of model 123114912.

TABLE V
FEATURE IMPACT RANKINGS

Rankings	Features
1	Acc_ped_ave
2	Insulation_res_ave
3	Insulation_res_std
4	Ave_charge_current
5	Alarm_times
6	Ave_charge_end_soc
7	Ave_charge_time
8	Ave_charge_start_soc
9	High_speed_rate
10	Ave_charge_soc_change
11	Delta_mile
12	Sum_mile
13	Insulation_res_abnor_rate
14	Month
15	Capacity
16	Speed_ave
17	Heavy_acc_ped_rate
18	Charge_segment_num
19	Ave_max_charge_current
20	Fast_charge_rate
21	Speed_std

REFERENCES

[1] X. Hu, L. Xu, X. Lin, and M. Pecht, "Battery lifetime prognostics," *Joule*, vol. 4, no. 2, pp. 310–346, 2020.

[2] K. A. Severson, P. M. Attia, N. Jin, N. Perkins, B. Jiang, Z. Yang, M. H. Chen, M. Aykol, P. K. Herring, D. Fraggedakis, M. Z. Bazant, S. J. Harris, W. C. Chueh, and R. D. Braatz, "Data-driven prediction of battery cycle life before capacity degradation," *Nature Energy*, vol. 4, no. 5, pp. 383–391, 2019.

[3] J. Zhu, Y. Wang, Y. Huang, R. B. Gopaluni, Y. Cao, M. Heere, M. J. Mühlbauer, L. Mereacre, H. Dai, X. Liu, A. Senyshyn, X. Wei, M. Knapp, and H. Ehrenberg, "Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation," *Nature communications*, vol. 13, no. 1, p. 2261, 2022.

[4] Y. Xing, C. Lv, X. Mo, Z. Hu, C. Huang, and P. Hang, "Toward safe and smart mobility: Energy-aware deep learning for driving behavior analysis and prediction of connected vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 7, pp. 4267–4280, 2021.

[5] M. Jafari, A. Gauchia, K. Zhang, and L. Gauchia, "Simulation and analysis of the effect of real-world driving styles in an EV battery performance and aging," *IEEE Transactions on Transportation Electrification*, vol. 1, no. 4, pp. 391–401, 2015.

[6] Z. He, X. Shen, Y. Sun, S. Zhao, B. Fan, and C. Pan, "State-of-health estimation based on real data of electric vehicles concerning user behavior," *Journal of Energy Storage*, vol. 41, p. 102867, 2021.

[7] H. Liu, Z. Deng, Y. Yang, C. Lu, B. Li, C. Liu, and D. Cheng, "Capacity evaluation and degradation analysis of lithium-ion battery packs for on-road electric vehicles," *Journal of Energy Storage*, vol. 65, p. 107270, 2023.

[8] D. Zhang, Z. Wang, P. Liu, C. She, Q. Wang, L. Zhou, and Z. Qin, "A multi-step fast charging-based battery capacity estimation framework of real-world electric vehicles," *Energy*, vol. 294, p. 130773, 2024.

[9] G. Pozzato, A. Allam, L. Pulvirenti, G. A. Negoita, W. A. Paxton, and S. Onori, "Analysis and key findings from real-world electric vehicle field data," *Joule*, vol. 7, no. 9, pp. 2035–2053, 2023.

[10] Shanghai Electric Vehicle Public Data Collecting, Monitoring, and Research Center, https://www.shevdc.org.

[11] Z. Deng, L. Xu, H. Liu, X. Hu, Z. Duan, and Y. Xu, "Prognostics of battery capacity based on charging data and data-driven methods for on-road vehicles," *Applied Energy*, vol. 339, p. 120954, 2023.

[12] H. Wang and J. Leskovec, "Combining graph convolutional neural networks and label propagation," *ACM Transactions on Information Systems*, vol. 40, no. 4, pp. 1–27, 2021.

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